Power Series and Taylor Series

ANSWER KEY

Section 1: Radius and Interval of Convergence

1 Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $R = 3$ ✓

b. $R = 1/3$

c. $R = 1$

d. $R = \infty$

Ratio Test: $\lim_{n\to\infty} \left| \frac{(x-2)^{n+1}}{(n+1)3^{n+1}} \cdot \frac{n3^n}{(x-2)^n} \right| = \frac{|x-2|}{3} < 1 \Rightarrow |x-2| < 3$. Thus $R = 3$.

2 Find the exact interval of convergence for $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n \cdot 3^n}$.

a. $[-1, 5)$ ✓

b. $(-1, 5)$

c. $[-1, 5]$

d. $(-1, 5]$

Center $c = 2, R = 3 \Rightarrow (-1, 5)$. At $x = -1$: $\sum \frac{(-1)^n}{n}$ converges by AST. At $x = 5$: $\sum \frac{1}{n}$ diverges (harmonic). Interval: $[-1, 5)$.

3 Find the radius of convergence of $\sum_{n=0}^{\infty} n!\,(2x)^n$.

a. $R = 0$ ✓

b. $R = 1/2$

c. $R = 1$

d. $R = \infty$

Ratio: $\lim_{n\to\infty} \frac{(n+1)!\,|2x|^{n+1}}{n!\,|2x|^n} = \lim_{n\to\infty} (n+1)2|x| = \infty$ for any $x \neq 0$. Thus $R = 0$ (converges only at $x = 0$).

Section 2: Geometric Series Representations

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4 Find the power series representation for $f(x) = \frac{1}{1+2x}$ centered at 0.

a. $\sum_{n=0}^{\infty} (-1)^n 2^n x^n$ ✓

b. $\sum_{n=0}^{\infty} 2^n x^n$

c. $\sum_{n=0}^{\infty} (-1)^n x^n$

d. $\sum_{n=1}^{\infty} (-2x)^n$

Using $\frac{1}{1-r} = \sum_{n=0}^{\infty} r^n$ with $r = -2x$: $\frac{1}{1-(-2x)} = \sum_{n=0}^{\infty} (-2x)^n = \sum_{n=0}^{\infty} (-1)^n 2^n x^n$ for $|x| < 1/2$.

5 Find the Maclaurin series for $\frac{x}{4-x^2}$.

a. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}$ ✓

b. $\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n}$

c. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}$

d. $\frac{1}{4}\sum_{n=0}^{\infty} x^{2n+1}$

$\frac{x}{4(1-x^2/4)} = \frac{x}{4}\sum_{n=0}^{\infty} (x^2/4)^n = \frac{x}{4}\sum_{n=0}^{\infty} \frac{x^{2n}}{4^n} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}$.

6 Find the Maclaurin series for $\ln(1+x)$ by integrating the series for $\frac{1}{1+x}$.

a. $\sum_{n=1}^{\infty} (-1)^{n-1}\frac{x^n}{n}$

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$

c. **Both of the above are equivalent** ✓

d. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$. Integrating term-by-term gives $\ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = \sum_{n=1}^{\infty} (-1)^{n-1}\frac{x^n}{n}$ for $|x| < 1$.

Section 3: Standard Maclaurin Series

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7 State the Maclaurin series for e^x.

a. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ ✓

b. $\sum_{n=1}^{\infty} \frac{x^n}{n}$

c. $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$

d. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

The standard Maclaurin series for $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$, valid for all $x \in \mathbb{R}$.

8 State the Maclaurin series for $\cos(x)$.

a. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$ ✓

b. $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$

c. $\sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$

d. $\sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n-1)!}$

Since $\cos(x)$ is an even function, its Maclaurin series contains only even powers: $\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots$.

9 Find the first three non-zero terms of the Maclaurin series for $\sin(2x)$.

a. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

b. $2x - 4x^3 + \frac{4}{3}x^5$

c. $2x - \frac{8}{6}x^3 + \frac{32}{120}x^5 = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

d. **Both of the above are correct** ✓

Substitute $2x$ into $\sin u = u - \frac{u^3}{6} + \frac{u^5}{120}$: $2x - \frac{8x^3}{6} + \frac{32x^5}{120} = 2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$.

Section 4: Taylor Polynomials and Lagrange Error Bound

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10 Find the 2nd-degree Taylor polynomial $T_2(x)$ for $f(x) = \sqrt{x}$ centered at $a = 4$.

a. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$ ✓

b. $T_2(x) = 2 + \frac{1}{4}(x-4) - \frac{1}{32}(x-4)^2$

c. $T_2(x) = 2 + \frac{1}{2}(x-4) - \frac{1}{16}(x-4)^2$

d. $T_2(x) = 2 + \frac{1}{4}(x-4) + \frac{1}{64}(x-4)^2$

$f(4) = 2$. $f'(x) = \frac{1}{2}x^{-1/2} \Rightarrow f'(4) = 1/4$. $f''(x) = -\frac{1}{4}x^{-3/2} \Rightarrow f''(4) = -\frac{1}{32}$.
$T_2(x) = 2 + \frac{1}{4}(x-4) + \frac{-1/32}{2!}(x-4)^2 = 2 + \frac{1}{4}(x-4) - \frac{1}{64}(x-4)^2$.

11 State the formula for the Lagrange Error Bound for a Taylor polynomial $T_n(x)$ centered at a.

a. $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$, **where** $M = \max|f^{(n+1)}(t)|$ **on** $[a, x]$ ✓

b. $|R_n(x)| \le \frac{M}{n!}|x-a|^n$

c. $|R_n(x)| \le M|x-a|^{n+1}$

d. $|R_n(x)| \le \frac{f^{(n+1)}(a)}{(n+1)!}|x-a|^{n+1}$

Taylor's Remainder Theorem states that the truncation error satisfies $|R_n(x)| \le \frac{M}{(n+1)!}|x-a|^{n+1}$ where M bounds the $(n+1)$th derivative on the interval between a and x.

12 Use the 3rd-degree Maclaurin polynomial for e^x to bound the error in estimating $e^{0.1}$. (Take $e^{0.1} < 3$).

a. $|R_3(0.1)| \le \frac{3}{24}(0.1)^4 = 1.25 \times 10^{-5}$ ✓

b. $|R_3(0.1)| \le 10^{-4}$

c. $|R_3(0.1)| \le \frac{1}{6}(0.1)^3$

d. $|R_3(0.1)| \le 10^{-6}$

$|R_3(0.1)| \le \frac{M}{4!}(0.1)^4$. Since $M = \max_{t \in [0,\, 0.1]} e^t = e^{0.1} < 3$, $|R_3| \le \frac{3}{24}(0.0001) = \frac{1}{8}(10^{-4}) = 1.25 \times 10^{-5}$.

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